

Computer vision

for histogram analysis of grape berry clusters

Mini Project

for

Current Applications of GIS GG5563

by Liron Chen, University of Aberdeen, April 2020

Index

Introduction

- Precision Viticulture
- Research
- Developments
- Technology
- Machine Vision
- Grape ripening
- Computer vision project

Methods

- Histogram equalisation
- Image segmentation
- RGB filtering
- Masking and histogram generation

Results and discussion

References

Appendix 1: R script

Appendix 2: result figures

Introduction

Precision viticulture

Precision viticulture seeks to improve the yield and quality of vineyard production by more effective management of inputs like energy, fertiliser, chemicals and labour, and reduce environmental impact. Detailed information can support high-resolution decision-making which avoids the inefficiency of a broad approach; variable rate machinery can automate locally detailed inputs and interventions, and the yield quality and quantity can be monitored and directed.

Vineyard production is determined by the the soil, climate, plant variety, water management, biotic stresses, and management. High variability can exist across a vineyard both spatially and temporally. GIS offers tools to integrate measurements and existing data, generate insight into growing conditions, and quantify management and input requirements, as well as support post-harvest processing such as mixing grapes to attain the desired wine quality.

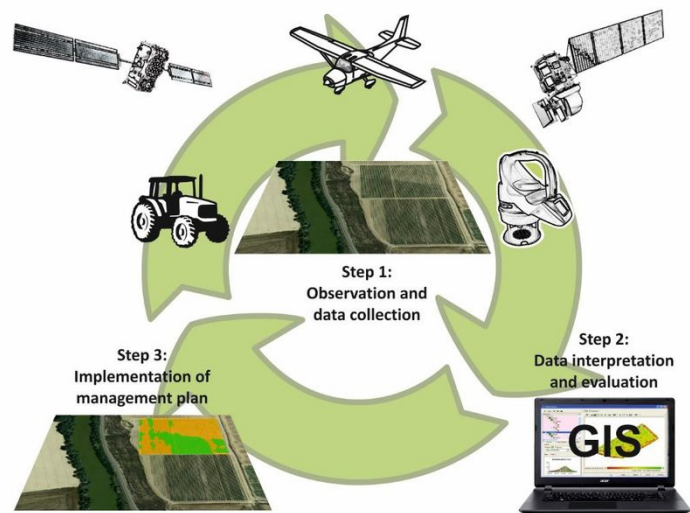


Figure 1: a conceptual management cycle for precision viticulture

Research

In the EU, vines were grown on 3.2 million hectares in 2015, 62% of the worlds wine was produced in the EU in 2018, while the EU wine exported over national borders amounted to 21.9 billion euros in 2017 (European Commission, 2017). The biggest importer was the UK.

Viticulture is a major source of income for countries in the EU; and subsequently much research and development has been undertaken and subsidised to optimise production.

Studies in precision viticulture in particular have been increasing rapidly over the past decade, as can be seen in figure 2. Of the most recent 50 studies, 35 directly involved remote or local sensing of vines and data analysis.

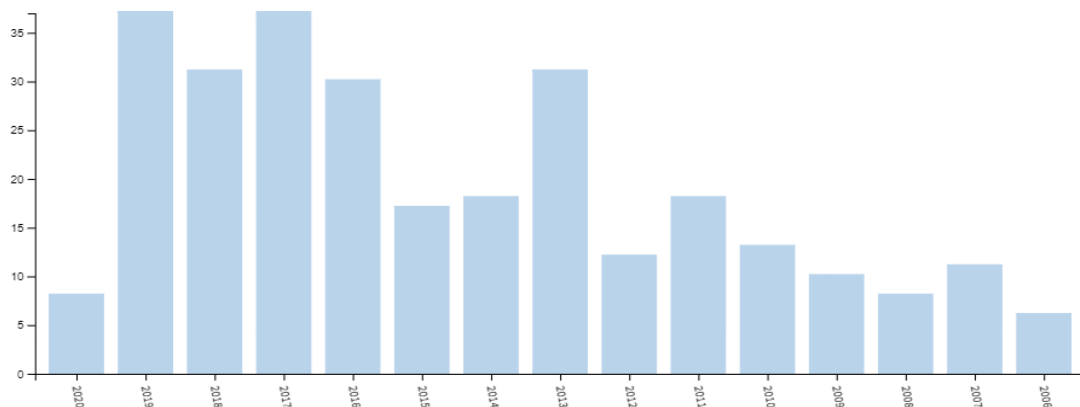
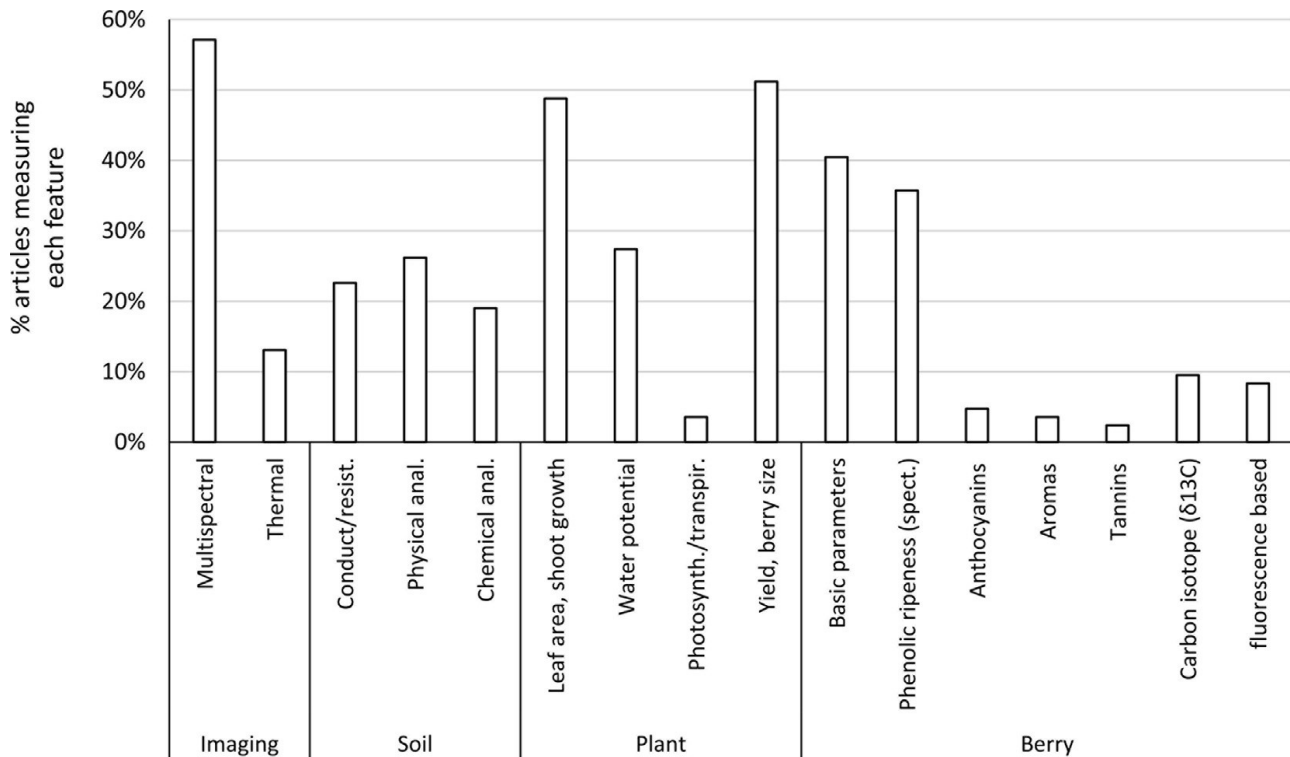


Figure 2: peer-reviewed publications in Precision Viticulture

Figure 3 shows some key concepts in PV studies (Santesteban, 2019). Note that these topics are not mutually exclusive; a study of shoot growth could be combined with thermal imaging and water potential measurements.

It is clear that multispectral imaging, leaf area and shoot growth, berry size and yield, phenolic ripeness are the most used variables. These focus on the most direct method to measure vine developments.



Relative presence of plant, soil and berry-related variables measured in research articles in Precision Viticulture from 2008–2017 (Santesteban, 2017)

Development

New technologies are constantly being implemented to support viticulture management.

Remote sensing data such as RGB imaging, multi- and hyperspectral sensing and laser surveying from satellites, aircraft and UAVs is used to examine plant physiology through concepts like NDVI (Normalised Difference Vegetation Index), LAI (Leaf Area Index) and PAB (Photosynthetically Active Biomass) (Matese and Gennaro, 2018) (Hall, 2002).

It can also be acquired from local, moving platforms like tractors, harvesters or mobile robots, which can collect higher resolution data such as 3D reconstructions of individual plants or direct measurements of berry cluster sizes.

Hand held equipment such as spectrophotometers can assess grape quality and maturation through measurements of sugar, acidity and anthocyanin concentrations and water content of individual berry clusters.

Fixed sensors that record micrometeorological conditions from soil to canopy, soil pH and nutrient levels, or sap flow through the plant have become more accessible with the reduced costs of wireless technology has become affordable.

The yield can be analysed during and after harvests to verify non-destructive measurements and predictions.

Combined with known factors such as grape variety, geology, fertiliser input and irrigation, GIS software can process the sensor data into a high quality model of the growth within the vineyard, which can reveal plant health, stresses, deficiencies, pests, and suggest optimal spatial distribution and timing for operations such as pruning, fertilisation, spraying, defoliation and harvesting, and can be exported to prescription maps.

Technology

In large vineyards, much of the work is mechanised. With precision viticulture however, mechanisation needs to be able to accommodate for locally varied operations and be able to work from prescription maps. Variable Rate Technology is a term for mechanisation that is able to do that, such as tractors, harvesters and sprayers that can target or skip over individual plants (Matese and Gennaro, 2018).

Data collection is also done by dedicated machinery which is typically much smaller than agricultural mechanisation. Examples of these are VINBOT, VineRobot and Wall-Ye; these machines are autonomous and due to their complexity are described as agricultural robots.

The line between VRT machinery and robotics is fading; many larger machines are equipped with sensors and real-time processing to direct operations, while many smaller agricultural robots are able to perform tasks such as pruning or harvesting in the case of Wall-Ye. The agricultural robots sector is projected to see a compound annual growth rate of over 20% for the next 5 years (MarketsandMarkets, 2019) and has been heralded as the future of farming (Dunlop, 2017).

Multiple types of robots for measurements, pruning, harvesting, and integrated arrays of operations have been developed and are actively used. They will need continued development and they imply that farming will become a more technology centred profession; yet the benefits are unmistakeable and could carry over into other sectors of agriculture.

Machine vision

Many of the agricultural robots, both the operational type such as harvesters or defoliators as the investigative and monitoring types such as VINBOT or VineRobot, use a type of machine and computer vision to examine the grape vines and berry clusters. They rely on multi- or hyperspectral sensors, RGB cameras, and a variety of other instruments. Identification of the vines and berry clusters is automated with machine vision, while the processing of the data to extract information about the characteristics of the plants and fruits is often done by automated or supervised computer vision (Tardáguila, 2012).

Grape ripening

Grapes initiate the process of ripening when the seeds are developed and the grape is made attractive to seed-dispersing animals. This ripening phase is called 'veraison', a period of chemical and physiological changes in the grape .

During this period, the grape transforms from hard, green, acidic and astringent, to aromatic, sweet, often darkly coloured, and mildly acidic. The grape reduces its photosynthetic activity and starts accumulating sugars, proteins, flavour and aroma compounds, and in red grapes, anthocyanins (Kennedy et al, 2007).

Anthocyanins are red and give the red grape and red wine its distinctive colour. Although they are only an indication of the ripeness of the grape, their production pathway is very similar to that of tannins and both of these compounds are heavy influences in phenolic ripening.

Computer visions project

This report will detail the creation of an automated tool in R to analyse images for the presence of berry clusters, and extract histograms for the different colour channels, to be used as index values for the assessment of ripeness. It is intended as an example of the programming of a vineyard robot that travels through the rows and sends information about the stages of ripeness to the farmers computer.

Methods

In the absence of multi- or hyperspectral data, RGB, images of grapevines with ripening clusters have been collected from google.

A script has been written in R to analyse the images and extract a histogram of the three channels. The full script can be seen in Appendix 1.

The script relies on four main processes. First, the histogram is equalised. Second, image segmentation is performed using SLICO superpixels and affinity propagation clustering. Thirdly, a pass is performed that filters through 6 custom RGB ranges to extract only the berry cluster areas. Finally, the cluster area is used as an inverse mask on the original image to extract only the berries and a histogram is generated for them. Figure 1 shows each stage of the process using an initial photograph on which the script was modelled. Appendix 2 shows 12 more photographs that were analysed with the same script.

Histogram equalisation

A histogram equalisation was performed on the image with the 'Imager' R package to increase contrast and improve further processing. Histogram equalisation increases or decreases the intensity (amount of light) of each pixel so that for each level of intensity there are an equal number of pixels in the image. This can be visualised as a redistribution of pixels in a grayscale histogram to fill all bars equally. The hue of pixels is not altered in this technique. Subsequent methods rely on similarity in pixel values, but the analysis mainly depends on colour and not light levels. Therefore, redistributing intensity levels will prevent overrepresentation from large groups pixels in small intensity ranges.

The result of histogram equalisation on the photograph can be seen in the second panel in figure 1.

Image segmentation

The enhanced image was then put through a superpixel image segmentation function (Zhou, 2015). The function first overlays the image with a grid of superpixels (in this script 600), adjusts their boundaries by tracing large changes in colour and light which we see as natural boundaries, and assigns the superpixel the value that has the greatest similarity to the cluster it covers. It then uses affinity propagation clustering on the superpixels, which

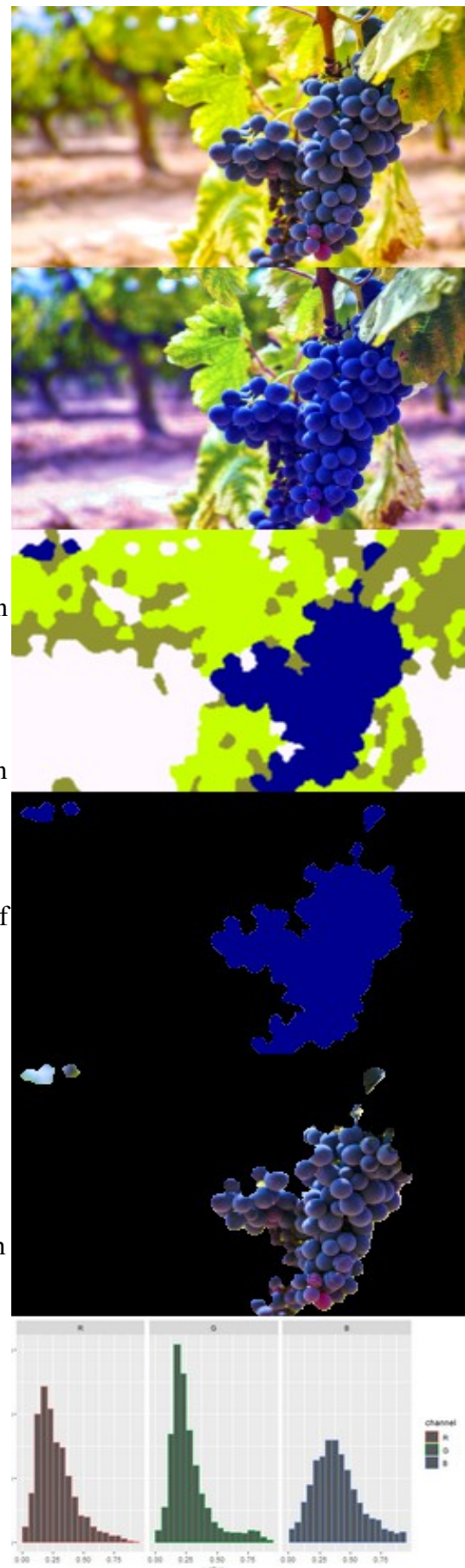


Figure 1: the initial grapevine photograph, the effects of the various stages of processing on it, and the histograms of the three colour channels of the berry cluster.

compares pairs of data points for similar characteristics until groups of superpixels are found that are closely related around a central value, and combines those into regions. The advantage using this combination of techniques is that it reduces the amount of data points for the APC technique to just 600, which speeds up the process immensely. The superpixelisation simplifies the image to the most unique values and boundaries, but for this project that is constructive. The result can be seen in the third panel of Figure 1.

RGB filtering

This image was then filtered over 6 custom ranges of RGB values, which were determined by examining a 3-dimensional RGB colour space for purple tones. An interactive model was found in the digital book Programming Design Systems website (Madsen) and can be seen in figure 2. The ranges used are shown in table 1. The result of the filtering can be seen in the fourth panel of Figure 1.

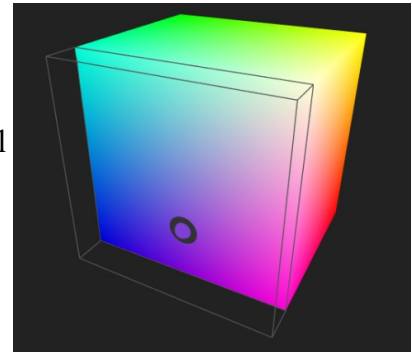


Figure 2: a view of the interactive 3D model of RGB colour space. The currently selected colour is R125 G65 B215.

	Red	Green	Blue
Range 1	0 – 190	0 – 127	229 – 255
Range 2	0 – 190	0 – 102	178 – 229
Range 3	0 – 190	0 – 76	127 – 178
Range 4	0 – 178	0 – 76	102 – 127
Range 5	0 – 127	0 – 38	63 – 102
Range 6	0 – 76	0 – 12	38 – 63

Table 1: six ranges of RGB values for filtering of the segmented image.

Masking and histogram generation

The resulting image was then used as a mask on the original photo to extract the image seen in the fifth pane of Figure 1. A histogram was calculated for each colour channel and saved, and can be seen in the bottom panel.

Results and discussion

The images resulting from the analysis of 12 more images can be seen in appendix 2. The script is not able to see grapes obscured by leaves, for yield estimates it would be best if multiple photos from different angles for each cluster could be combined.

Most images were processed to reasonable accuracy, although higher resolution preprocessing could remove the slivers of leaf and other elements that have been included in the superpixels and will affect the histograms. Images 8 and 11 have not been processed correctly, and 10 and 13 do not have complete cover.; this is likely due to the colour definition of the RGB filter, as the process functioned correctly up to that stage.

It is likely that this can be corrected by a more expansive set of preprocessing, the use of multiple script based on a preliminary histogram analysis, or an adjustment of the RGB ranges.

However, using hyperspectral data would be even better: that will remove the errors in patches of sky or stems being identified as grapes, it will allow for the correct identification of white grapes as well, and make implementation of the PROSPECT and PROSAIL models to analyse grape chemistry and physiology and water stress possible.

The model can also be adjusted to analyse leaf properties, which would also benefit from wide-

spectrum data and more advanced models.

The histograms produced are only a starting point for ripeness analysis, the details of anthocyanin production and their relationship to phenolic ripeness will have to be quantified for the histogram values to represent information. The connection exists however and an approximation of ripeness should already be feasible with the generated images.

Finally, the cluster weight can be calculated from the pixel surface area if the distance to the berry cluster is known, or multiple photos of the same cluster can be taken at different distances (Atherton et al, 1991). However, with the available data, this was not possible.

The script as an example of automated information extraction from RGB imagery functions, and is a basic example of remote sensing computer vision that could be used for many different applications.

References

Atherton, T.J., Kerbyson, D.J. and Nudd, G.R., 1991. Passive estimation of range to objects from image sequences. In BMVC91 (pp. 343-346). Springer, London.

Kennedy, J., Robinson, S. and Walker, M., 2007. Grape and wine tannins: Production, perfection, perception. Practical Winery and Vineyard, pp.57-67.

Matese, A. and Di Gennaro, S.F., 2015. Technology in precision viticulture: A state of the art review. Int. J. Wine Res, 7, pp.69-81.

Santesteban, L.G., 2019. Precision viticulture and advanced analytics. A short review. Food chemistry, 279, pp.58-62.

Tardáguila, J., Diago, M.P., Millan, B., Blasco, J., Cubero, S. and Aleixos, N., 2012, June. Applications of computer vision techniques in viticulture to assess canopy features, cluster morphology and berry size. In I International Workshop on Vineyard Mechanization and Grape and Wine Quality 978 (pp. 77-84).

Zhou, Bao. "Image segmentation using SLIC superpixels and affinity propagation clustering." Int. J. Sci. Res 4.4 (2015): 1525-1529.

https://ec.europa.eu/eurostat/statistics-explained/index.php/Vineyards_in_the_EU_-_statistics

<https://www.marketsandmarkets.com/Market-Reports/agricultural-robot-market-173601759.html>

<https://www.theguardian.com/sustainable-business/2017/may/09/agbots-next-gen-farming-and-how-they-can-teach-us-about-the-future-of-work>

<https://programmingdesignsystems.com/color/color-models-and-color-spaces/index.html>